

# The training system

What to do, how often, and how well the evidence backs it.

| THE WEEK                                     |                                                            | A strong     | B mechanistic | C convention |
|----------------------------------------------|------------------------------------------------------------|--------------|---------------|--------------|
| <b>Max static holds</b>                      | Full relaxed inhale, lie still, breathe only when you must | 2 days · 5–8 | A             |              |
| <b>CO<sub>2</sub> / O<sub>2</sub> tables</b> | Timed holds with set rests — trains the urge to breathe    | 2–3 days     | B             |              |
| <b>IMST</b>                                  | 30 hard inhales through a handheld resistance trainer      | most days    | A             |              |
| <b>Rest or stretch</b>                       | Gentle ribcage & side-body stretches, or a full day off    | 1 day        | B             |              |
| <b>Apnea walk</b>                            | Walk while holding, after a normal breath out              | optional     | C             |              |

| A MAX HOLD, STEP BY STEP |                                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1                        | <b>Settle</b> — 1–2 min of slow, easy breathing. Breath out a little longer than in. Never fast or deep.                |
| 2                        | <b>Fill</b> — one full but relaxed inhale, to comfortable. No gulping extra air.                                        |
| 3                        | <b>Hold</b> — go slack. Easy at first, then diaphragm contractions in waves. Relax through them — that is the training. |
| 4                        | <b>Stop</b> — contractions won't settle, or any dizziness, tingling or narrowing vision. Then 3 hook breaths.           |

| PROGRESSION |           |                                                                                |
|-------------|-----------|--------------------------------------------------------------------------------|
| 2 weeks     | +30–45 s  | Fast — CO <sub>2</sub> tolerance and a stronger dive reflex, not new capacity. |
| 3 months    | 2:00–2:30 | Real metabolic adaptation begins.                                              |
| 3–6 mo      | 2:30–3:30 | Realistic for most healthy adults training daily.                              |
| 6 mo+       | slow      | Gradual, as you approach your personal ceiling.                                |

Population averages. Your lungs, spleen, chemoreceptors and body size set a personal limit the research cannot predict for you.

**Plateau** — four weeks with no new best → a 1–2 week deload: recovery and stretching, not more volume. **Retest** the max every 2–4 weeks and reset the table targets.

| NEVER                                  |                                    |
|----------------------------------------|------------------------------------|
| Hyperventilate before a hold           | Pack your lungs                    |
| Breath-hold in water without a spotter | Wim Hof breathing in or near water |

# Doing a hold

On land, lying down, with someone in the room. Never in water alone.

## 1 · BEFORE THE HOLD

**Settle.** 1–2 minutes of slow, easy breathing — a gentle breath in, a slightly longer breath out (about in for 4, out for 6). You are calming down, not stocking up on air.

**Last breath.** One full but relaxed inhale. Fill to comfortable, then stop — no gulping or packing extra air.

Never breathe hard and fast to “prepare”. It clears the CO<sub>2</sub> that would warn you, and that is what causes blackout.

## 2 · DURING THE HOLD

Lie still. Let your face, jaw, shoulders and hands go completely slack.

The first stretch is easy — no urge yet. Then your diaphragm starts pulling in waves. That is normal, and it is the part that trains you. Keep relaxing through it.

## 3 · END THE HOLD WHEN

The contractions are strong, constant and you cannot relax them — or the moment you feel dizzy or tingly or your vision narrows. Never push past that. A samba (loss of motor control) after a hold means stop for the day.

## 4 · RECOVER

Straight after: three hook breaths — a quick full inhale, hold it a second against a slightly closed throat, let it fall out. Then breathe normally.

## THE SETS

|                       |                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| Max holds             | 5–8 holds · 2–3 min of normal breathing between each                                                                  |
| CO <sub>2</sub> table | Same hold length every round ( $\approx$ half your max) · rests shrink $\sim$ 15 s per round · 6–8 rounds             |
| O <sub>2</sub> table  | Hold gets longer each round (toward 80% of max) · rest fixed at $\sim$ 2 min                                          |
| IMST                  | 30 hard inhales through a handheld resistance trainer · start $\sim$ 50%, build to 75%                                |
| Stretch               | Gentle chest, side and upper-back stretches — breathe a full easy breath into each. Loosens the ribcage; not packing. |

Never hyperventilate. Never breath-hold alone in water.

# What to expect

From a ~1:30 start, training daily on land. Static hold, not in water.



|            |           |                                                                                                                                                    |
|------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 weeks    | +30–45 s  | The big, fast jump. Almost all of it is CO <sub>2</sub> tolerance, a stronger diving reflex, and learning to sit with the urge — not new capacity. |
| 1 month    | climbing  | Still going up, slower now, and still all tolerance.                                                                                               |
| 3 months   | 2:00–2:30 | Real metabolic adaptation appears — less acidosis and oxidative stress after a hold (Joulia 2003).                                                 |
| 3–6 months | 2:30–3:30 | A realistic target for most healthy adults training daily.                                                                                         |
| 6 months+  | slow      | Progress keeps coming, just gradually, as you close on your own ceiling.                                                                           |

READ THIS PART

These are population averages. Your lung size, spleen size, chemoreceptor set-point and body size set a personal limit — and the research cannot tell you what yours is. Some people pass 3:30 in months; some level off below it and are training correctly.

Dynamic apnea — holding while swimming — runs well below your static time, because moving burns oxygen faster.